

Heel-Toe Tap: A Mode Switching Gesture for Engaging Foot-Assisted Gaze Selection and Reduced Unintentional Inputs

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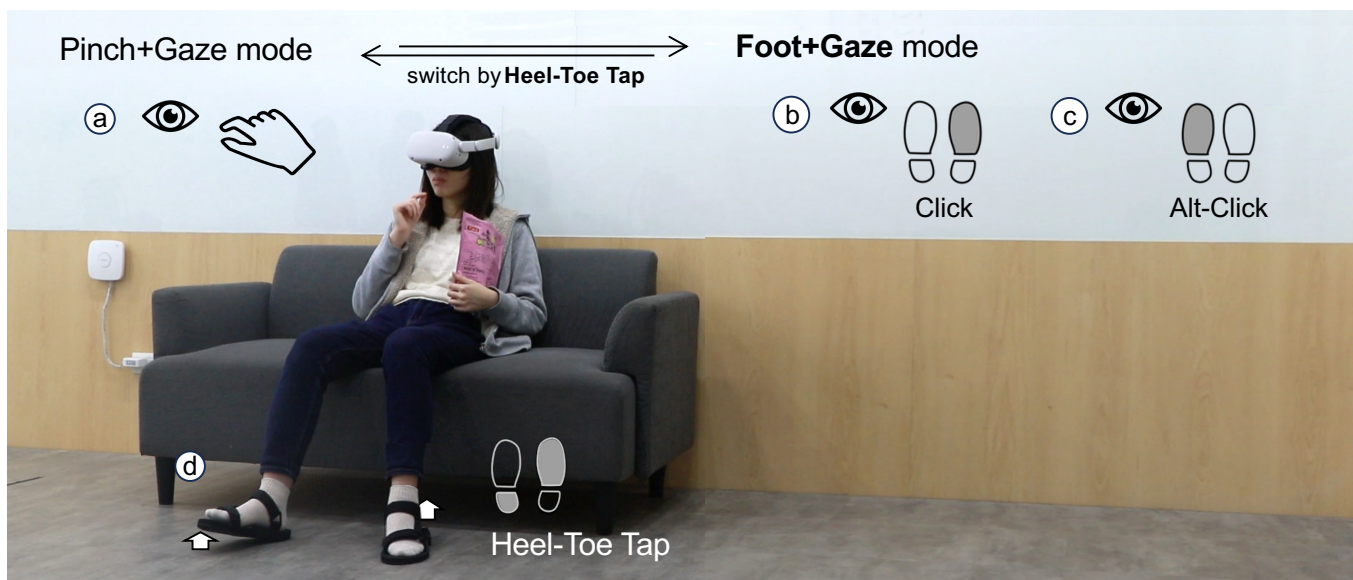


Figure 1: The user switches from (a) Pinch+Gaze input to (bc) Foot+Gaze input (d) by performing a diagonal heel-toe tap with the feet. (bc) In the Foot+Gaze mode, a tap with the dominant foot triggers a click input at the gaze cursor, while a tap with the non-dominant foot sends an alt-click input.

Abstract

Gaze combined with finger pinching is an emerging input method for head-mounted displays. However, in situations where the hands are occupied, foot tapping offers a practical alternative for gaze-based selection. A key challenge in employing foot tapping for gaze-based selection is its susceptibility to unintended inputs, largely due to the foot's resting position on the ground. Minor, unintentional movements can be misinterpreted as deliberate taps. To address this issue, we focus on designing switch gestures that activate "foot

mode," to prevent accidental activations. We introduce a diagonal heel-top tapping method as an effective switch gesture, which minimizes unintentional activations while maintaining ease of use. Additionally, we demonstrate an application that allows seamless switching between Pinch+Gaze and Foot+Gaze interactions, utilizing the proposed switch gesture.

CCS Concepts

• Human-centered computing → Gestural input.

Keywords

Foot gesture, in-place foot input, seated interaction, foot-based interaction

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1 Introduction

Gaze combined with finger pinching, or Pinch+Gaze interaction, is a promising input method for head-mounted displays (HMDs), where the recent the Apple Vision Pro is a prominent example. In situations where the hands are unavailable—like when someone is holding a bag of chips—foot tapping offers an effective alternative to the pinch gesture to assist gaze-based selection. However, foot tapping is prone to unintended inputs, as the default resting position of the foot is on the ground. Any minor movement that involves repositioning the foot can be misinterpreted as a tapping action.

The challenge in designing a Foot+Gaze interaction system lies in accurately distinguishing intentional foot taps meant for input from everyday foot motions. Two approaches can address this issue. The first is automatic detection of intended foot taps. However, this is challenging because accidental foot taps can occur in many situations, and unreliable detection may result in erroneous gaze-based selections. The second approach involves using a manual switch to activate “foot mode,” where only foot motions during this mode are recognized as input. This allows users to clearly define when they intend to use foot taps for interaction versus when they want to relax without triggering accidental inputs.

This study explores the design of mode switching gestures for activating the “foot mode” switch, aiming to enable mode switching via foot movements when transitioning from Pinch+Gaze to Foot+Gaze interaction, particularly when the user’s hands are unavailable. It is crucial to identify foot movements that are easy to perform and less likely to be triggered unintentionally. We chose foot tapping for the mode-switch design because it involves minimal leg movement, making it easy to execute without excessive effort. Previous research also indicates that foot tapping is less physically demanding compared to other movements, such as lateral foot motion or heel rotation [12]. To enhance the design, we included both heel and toe tapping and introduced the concept of “synchronous tapping.” Synchronous tapping involves coordinated tapping of both feet in specific patterns. As illustrated in Figure 2, we defined four types of synchronous tapping: , , , . Additionally, we incorporated double tapping, which consists of consecutive taps using the same part of the foot, including , , , . Double tapping is frequently used in foot-based input systems to provide an extra layer of confirmation for user actions. In total, we evaluated eight foot-tapping designs: four double-tap gestures and four synchronous tap gestures. These designs were assessed using objective measures such as completion time and success rate, as well as subjective metrics like physical demand, mental demand, and concern about accidental inputs.

Our results suggest that diagonal gestures are the most suitable for mode switching, as they are the least prone to accidental inputs. While diagonal gestures exhibited completion times and success rates comparable to non-diagonal gestures, they also imposed greater physical and mental demands. Finally, diagonal gestures involve using either the forefoot of the dominant or non-dominant

foot with , respectively. We recommend adopting the diagonal gesture that involves the user’s dominant foot, e.g.,  → , as it allows a smooth continuation to click with toe tapping with the dominant foot, as demonstrated in Figure 1.

It is important to note that Foot+Gaze input is not intended to replace the more precise Pinch+Gaze interaction. Hands and fingers offer a level of dexterity that feet cannot match. For example, Pinch+Gaze interactions allow for more complex actions, such as combining hand motion with a pinch to drag items in the interface, something that would be difficult to replicate with foot motions. Therefore, Foot+Gaze input is more suited to sparse tasks, particularly in situations where users prefer to relax their hands, such as while sitting on a sofa.

The contributions of the work are threefold. (1) This study is the first to highlight the importance of designing mode switching gestures for foot interaction, with the goal of preventing unintentional inputs. Specifically, we develop mode switching gestures that facilitate foot-assisted gaze selection for seated users. (2) Our study identifies the diagonal heel-toe tapping method as an effective mode switching gesture for foot interaction. (3) We showcase an application that allows users to quickly switch between Pinch+Gaze and Foot+Gaze interactions, utilizing the proposed mode switching gesture, thereby demonstrating its practical implementation.

2 Related Work

We review related works on the design of foot-assisted gaze interaction and mode switching gestures.

2.1 Foot-Assisted Gaze Interaction

Research on the usability and permanence of foot-assisted gaze interactions is relatively limited. Rajanna and Hammond introduced GAWSCHI [13] that uses foot tapping to confirm gaze selection. They demonstrated that this combination can be as effective as mouse-based interactions in terms of time and accuracy, provided that the interface elements are above a certain size threshold. Klamka et al. [8] combined gaze input with a foot pedal to perform secondary mouse tasks, such as panning and zooming, and found that gaze-supported foot input allows intuitive navigation, comparable to mouse input. Similarly, [2] explored the use of various foot postures for controlling panning and zooming, addressed by gaze input. Additionally, Hatscher et al. [4] enabled foot-step interaction on a remote monitor, where users control medical images during minimally-invasive procedures through gaze-directed input.

Surprisingly, only the GAWSCHI framework [13] explicitly introduced foot tapping for gaze selection, but it lacks a comprehensive analysis of the method. Other previous works have focused on integrating gaze selection with foot input for richer interactions, such as using foot postures [2, 4] to zoom and pan media or controlling pedal devices with the foot [8]. These studies consistently suggest that the foot is well-suited for gaze interaction.

However, none of the existing research addresses the issue of accidental foot input. While foot pedals [8] offer a clear mode-switching mechanism, they restrict user mobility. In contrast, foot gestures based on wearable sensing preserve mobility but are prone to unintentional activation in daily use. This study focused on enhancing gaze selection using foot tapping gestures, a gesture type

proven to be quicker to perform than other foot-based interactions but also more susceptible to accidental input. Our work is the first to emphasize the importance of developing a foot gesture that minimizes the risk of unintentional activation, making it suitable for toggling foot input in systems integrated with gaze selection.

2.2 Mode Switching Gestures

Mode switching gestures are specific gestures used to define the start and end of an input sequence, helping to prevent accidental or unintended inputs. Common examples include a designated swipe gesture to accept a call on smartphones and a voice command like "Hey Siri" to initiate further voice input. The primary goal of mode switching gestures is to enhance usability by ensuring that only intentional inputs are registered.

Mode switching gestures should be distinct from everyday movements, making them less prone to accidental activation. Various modalities of mode switching gestures have been explored in HCI research. For instance, in pen-based user interfaces, a long press can differentiate between inking and gesturing [5, 10]. Bimanual gestures involve triggering an action mode with the non-dominant hand, followed by performing a command with the dominant hand [15]. Device-specific trigger actions, such as DoubleFlip [14], which requires a large rotation, and WristRotate [7], which introduces a wrist rotation technique as the mode switch for smartwatches, have also been studied. Additionally, some techniques, like using the back of the index finger as a slider input or exploring rhythmic microgestures that involve compound finger movements, expand the possibilities for mode switching gestures. Introducing additional modalities, BlynSync [18] proposes synchronizing a blink with a touch gesture as a mode switch on smartwatches. Other efforts involve engaging end-users in the design process to reduce the risk of confusing gestures with natural movements [16, 19]. A common characteristic among these gestures is that they consist of motions that are not typical in everyday activities, thereby reducing the likelihood of accidental input. Moreover, other works have used machine-learning approaches to predict the false positive rates of gestures. These methods compare gestures against a database of everyday actions to identify those most robust to false activations [6, 9, 11]. Our work focused on studying mode switching gestures for foot motions, specifically within the context of Foot+Gaze interaction for seated users utilizing in-place foot taps.

3 DESIGN OPTIONS for Mode Gestures

Previous studies have explored various foot movements, such as tapping, translational actions, and rotational movements (e.g., heel rotation), as methods for device interaction [3, 17]. In this study, we focus on tapping as mode switching gestures for switching to the Foot+gaze input mode, due to its advantages over other foot movements. First, tapping requires less muscular effort than translational or rotational actions, which involve leg movement and the engagement of more complex muscle groups [3]. Second, tapping involves minimal leg movement, making it suitable for confined spaces and more socially acceptable in public settings [1]. Moreover, unlike translational or rotational movements that require repositioning (homing) after execution, tapping can be performed continuously without the need for resetting. Lastly, tapping-based

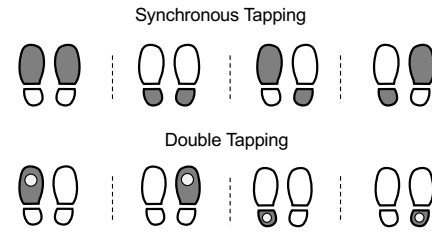


Figure 2: The eight design options for mode switching gestures: four types of synchronous tapping, which involve two concurrent single taps performed simultaneously across both feet, and four types of double tapping, which consist of two consecutive single taps using the same part of the foot.

mode switching gestures provide a coherent input vocabulary, allowing for seamless integration with regular toe-tapping-assisted gaze selection [4].

The potential of foot taps, incorporating both toe and heel tapping, has been harnessed to enable rich locomotion control for seated users [1]. In this study, we expand the foot tapping vocabulary by introducing synchronized tapping. We based on two tapping actions per foot: toe tapping and heel tapping. *Single tapping* refers to any of the four distinct tapping types performed individually across both feet. *Synchronous tapping* then involves executing two taps simultaneously, one with each foot, in a coordinated manner, as shown in Figure 2. During synchronous tapping, both feet remain grounded throughout the action, minimizing the engagement of leg muscles. Additionally, we added *double tapping* into the design options, which consists of consecutive taps using the same part of the foot. Double tapping is commonly used in foot-based input systems to provide an additional layer of confirmation for user actions.

4 USER STUDY

The objective of our study is to evaluate the previously discussed foot command patterns in terms of their objective performance, including completion time and success rates, as well as subjective experiences such as cognitive load and physical fatigue. We recognize that while some patterns may be easy to perform, they also carry a higher risk of unintentional activation. However, it is difficult to accurately measure this risk without real-world application. To address this, we asked participants to estimate the likelihood of accidentally triggering a pattern based on their own experiences, providing context for their responses.

4.1 Apparatus

For our technical setup shown in Figure 3, we implemented foot commands activated by toe and heel taps. We equipped each sandal with two infrared photoreflexive sensors (SG-105¹) on the front and back facing downward, following prior implementations [1]. These sensors detect taps up to 2 cm range, accommodating subtle movements and hesitations. A thresholding method was used to detect foot-up and foot-down states. The sensors were connected

¹SG-105 Photoreflexive Sensor: <https://datasheetspdf.com/pdf-file/487288/KODENSHIKOREACORP/SG-105/1>

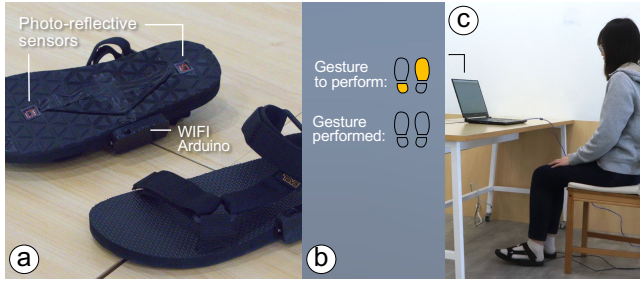


Figure 3: (a) The shoe prototype for detecting foot tapping. (b) The experimental setup: participants seated, wearing specially provided sandals, and executing designated tapping patterns in response to tasks displayed on the screen.

to a WIFI-ready Arduino mounted on the sandal, which wirelessly transmitted data to a PC. The study involved a custom Unity program on a Windows laptop, and participants, seated in non-swivel chairs, interacted with tasks and questionnaires displayed on an LCD screen.

4.2 Procedure and Task

The study was divided into two main parts. The first part quantitatively assessed performance through task completion times and error rates. This section included a practice session where participants familiarized themselves. It was followed by an official session consisting of four rounds, each involving eight random tapping patterns. Participants aimed to complete each pattern quickly and without errors. If a pattern was entered incorrectly, it was reattempted until successfully completed. The second part of the study qualitatively evaluated the participants' experiences. After each pattern attempt, participants assessed the physical and mental effort required and the potential for accidental activation using a 7-point Likert scale. Following a short break, a semi-structured interview was conducted to discuss participants' experiences and their ratings. In total, each participant spent about an hour completing the study.

4.3 Participants

Our study included 16 participants—7 females and 9 males—aged between 19 and 40 years (Mean: 24.1, SD: 4.4). Participants were recruited from the local university using snowball sampling techniques, ensuring they had no physical impairments and normal mobility in both lower limbs.

4.4 Results

Figure 4 presents the average completion time, and user ratings for the eight patterns, and Figure 5 the average ratings on physical demand, mental demand, and concern for accidental input.

4.4.1 Completion Time. The average completion time are 2.51 seconds (SD: 0.37) for LL , 2.54 (SD: 0.66) for LR , 2.81 (SD: 0.29) for RR , 2.90 (SD: 0.36) for RL , 2.91 (SD: 0.32) for LL , 2.88 (SD: 0.22) for LR , 2.91 (SD: 0.47) for RR and 2.89 (SD: 0.34) for RL . The ANOVA test revealed a significant difference on completion time, with a large

effect size ($F(6, 105) = 13.684, p < 0.001, \eta_p^2 = 0.477$). Bonferroni-corrected Post-hoc pairwise comparisons only found that both LL and LR were faster than the rest (all $p < 0.05$).

4.4.2 Success Rate. The average success rates are 96.0% (SD: 8.5%) for LL , 92.3% (SD: 12.5%) for LR , 100.0% (SD: 0.0%) for RR , 96.0% (SD: 8.5%) for RL , 100.0% (SD: 0.0%) for LL , 98.0% (SD: 6.3%) for LR , 100.0% (SD: 0.0%) for RR and 94.0% (SD: 10.1%) for RL . The results of the ANOVA showed no significant effect of the different conditions on success rates, $F(7, 105) = 1.345, p = 0.237, \eta_p^2 = 0.082$. Post-hoc comparisons using the Bonferroni correction did not reveal any significant differences between conditions ($p > 0.05$ for all comparisons).

4.4.3 Physical Demand. The average physical demand are 1.31 (SD: 1.46) for LL , 1.88 (SD: 1.20) for LR , 2.44 (SD: 0.89) for RR , 2.69 (SD: 1.29) for RL , 1.50 (SD: 0.48) for LL , 1.56 (SD: 0.77) for LR , 1.50 (SD: 1.36) for RR and 1.56 (SD: 1.08) for RL . The Friedman test revealed a significant difference in user ratings regarding the physical demand, with a moderate effect size indicated by Kendall's coefficient of concordance ($\chi^2(6, 16) = 43.780, p < .001, W = 0.456$). Post-hoc tests showed that one of the diagonal step RR caused more mental demand than all the other non-diagonal steps. The other diagonal step RL only caused more physical demand than for LL and LR (both $p < 0.05$).

4.4.4 Mental Demand. The average mental demand are 1.38 (SD: 1.26) for LL , 1.06 (SD: 1.66) for LR , 2.38 (SD: 1.09) for RR , 2.69 (SD: 1.08) for RL , 1.06 (SD: 0.25) for LL , 1.13 (SD: 0.00) for LR , 1.13 (SD: 0.00) for RR and 1.19 (SD: 0.00) for RL . The Friedman test revealed a significant difference in user ratings regarding the mental demand, with a notable effect size indicated by Kendall's coefficient of concordance ($\chi^2(6, 16) = 69.238, p < .001, W = 0.721$). Post-hoc tests showed that one of the diagonal step RR caused more mental demand than all the other non-diagonal steps (all $p < 0.05$). The other diagonal step RL also caused more mental demand than all non-diagonal steps except for LL and LR (all $p < 0.05$).

4.4.5 Concern of Accidental Input. The average concern of accidental input are 5.69 (SD: 2.21) for LL , 5.19 (SD: 0.72) for LR , 2.44 (SD: 1.55) for RR , 2.56 (SD: 1.46) for RL , 5.44 (SD: 1.36) for LL , 5.00 (SD: 1.84) for LR , 5.88 (SD: 1.84) for RR and 4.81 (SD: 2.34) for RL . The Friedman test revealed a significant difference in user ratings regarding the concern of accidental input, with a notable effect size indicated by Kendall's coefficient of concordance ($\chi^2(6, 16) = 60.610, p < .001, W = 0.631$). Post-hoc tests showed that both diagonal steps, RR and RL , were rated significantly lower in terms of concern for accidental input compared to all non-diagonal steps (all $p < 0.05$). However, no significant difference was found between the two diagonal steps, nor among the non-diagonal steps.

4.4.6 Discussion. We begin by addressing accidental input, a key consideration in our design. Diagonal steps consistently received lower ratings compared to non-diagonal steps, with participants frequently noting that these movements felt unfamiliar. Unlike commonly used foot motions in daily activities, diagonal steps do not rely on pre-existing muscle memory, which may explain

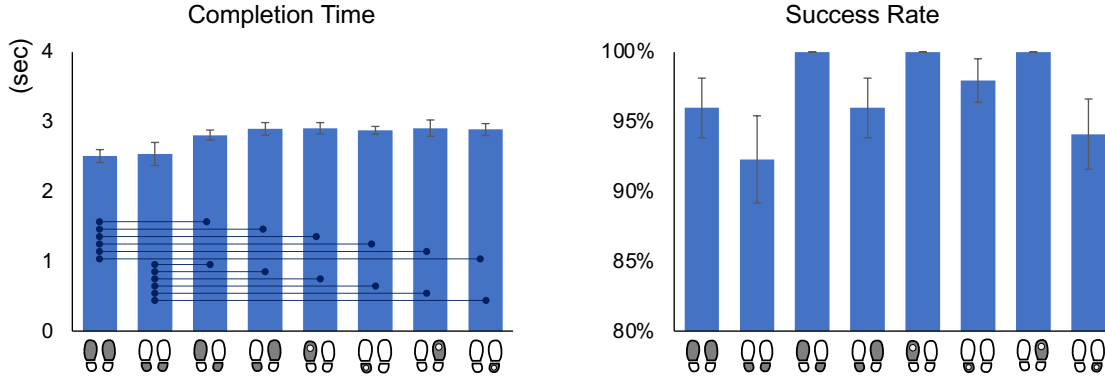


Figure 4: Results of completion time and success rate for the eight gestures. The error bars denote 1 standard error. Pairwise significant differences are indicated.

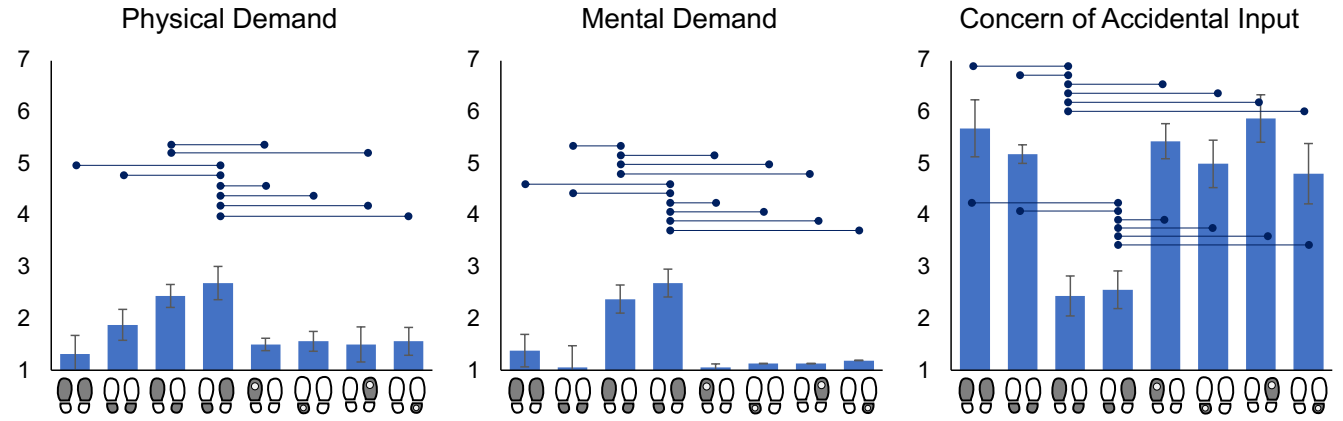


Figure 5: User ratings of physical demand, mental demand, and the level of concerns of unintentional activation for the eight gestures. The error bars denote 1 standard error. Pairwise significant differences are indicated.

the reduced concern for accidental activation when using these gestures.

However, this unfamiliarity also contributed to higher physical and mental demands compared to non-diagonal gestures. Participants reported that diagonal steps required practice to become comfortable, yet the effort required remained manageable, as indicated by average physical and mental effort ratings falling below the neutral threshold. This suggests that while diagonal steps introduce an initial learning curve, they do not pose a significant long-term usability barrier. This aligns with feedback such as: “The gestures were easy to learn and memorize.” Notably, despite the increased perceived effort, completion times for diagonal gestures were comparable to those of non-diagonal gestures, suggesting that users adapted relatively quickly.

When asked about their preference for adopting diagonal gestures to activate foot mode—where the dominant foot’s toe tapping serves as a click to assist gaze selection—participants overwhelmingly favored the diagonal gesture involving the forefoot of their dominant foot (e.g., → for right-foot-dominant users). They cited its ergonomic advantage and quicker transition to tapping

after performing the gesture. This preference highlights the importance of gesture fluidity, where the transition from mode switching to interaction should feel natural and efficient. As a result, we recommend the diagonal gesture, , for users with right-foot dominance and the second diagonal gesture, , for those with left-foot dominance, ensuring an intuitive and efficient interaction method tailored to individual preferences.

5 Example Application

We present the Foot+Gaze interaction technique for controlling appliances in a virtual smart environment, as illustrated in our Unity-implemented setup. This environment includes appliances like a TV, lamp, and fan, controllable via Pinch+Gaze or Foot+Gaze techniques when foot mode is activated, using a shoe prototype and an HTC VIVE Pro Eye headset with gaze tracking.

The interaction defaults to Pinch+Gaze mode. Foot mode is activated by a diagonal step with the right foot (assumed dominant), displaying a foot indicator on the cursor to signify activation. In this mode, toe-tapping the dominant foot executes a “click” on the



Figure 6: The remote control of interactive appliances begins in Pinch+Gaze mode and can switch to Foot+Gaze when the user's hands are occupied. This switch is made through a diagonal heel-toe tap. In Foot+Gaze, the user turns on the TV with a dominant foot tap and accesses a context menu with a non-dominant foot tap. She selects 'next video' with another dominant foot tap. To revert to Pinch+Gaze, she performs the same diagonal step and can relax with foot motions.

targeted appliance, toggling its primary function, such as turning on or off. Toe-tapping the non-dominant foot brings up a context menu for additional options like channel switching or brightness adjustment, depending on the appliance. To select an option from the context menu, the user gazes at the desired choice and taps the dominant foot. Tapping the non-dominant foot outside the menu area closes the menu, and any open menus close automatically when a new one is accessed. To revert to Pinch+Gaze mode, the user performs the same diagonal step, deactivating foot input. Now, consider a scenario where a user, seated with a snack, uses the system. She activates foot mode to keep her hands free, uses her dominant foot to turn on the TV, and accesses the TV menu with her non-dominant foot to change channels. She then interacts with the fan and lamp similarly, effectively managing her environment while enjoying her snack in a relaxed setting.

6 Conclusion

This study investigated reducing unintended inputs in foot-based interactions for seated users using gaze-controlled systems. We identified diagonal tapping as a reliable method to activate "foot mode," minimizing accidental inputs while maintaining usability. This facilitates seamless transitions between Pinch+Gaze and Foot+Gaze interactions, which is advantageous when users' hands are occupied. However, our assessment relied on subjective ratings to gauge accidental inputs, identifying diagonal heel-toe tapping as a promising method. To confirm the effectiveness of these gestures and their practical application, a longitudinal study is crucial.

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